

Lab 2: Community Detection in Networks

Step 1. Choose a network

- Pick a network dataset (≥ 1000 nodes and edges) from the [ICON index](#) or any other source of your interest.
 - Choose one where you understand what nodes and edges represent, so you can interpret communities meaningfully.
 - You may reuse the network you worked on in Homework 1.
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Step 2. Explore community detection

- Apply **at least two different community detection algorithms** of your choice (e.g., modularity-based, clique-based, spectral, Louvain, Leiden, label propagation, etc.).
 - You can use **NetworkX** or any other network analysis tool you prefer (e.g., igraph, graph-tool, Gephi).
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Step 3. Analyze results

For each algorithm:

1. Report the **number of communities** you find.
 2. Plot the **distribution of community sizes** (a histogram or complementary CDF; log-log if needed).
 3. Comment briefly on the results:
 - Are there very large or very small communities?
 - Do the communities align with your domain knowledge about the network?
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Step 4. Compare algorithms

- Compare the results across the algorithms you used:
 - Did they find a similar number of communities?
 - Do the size distributions look similar?
 - For the top few largest communities, are they roughly overlapping or very different?

- Summarize your insights.
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 Deliverables:

- A short report (with plots) showing your analysis.
- Your comments interpreting the results based on both statistics and domain knowledge.